Seat No. 6 2 5 1 5 2

stitute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

rogramme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

atch: 2025-29

emester: I

ourse: Physics for Quantum Computing

ourse Code: 0705210103

ate: 08/12/2025 Maximum Marks: 30

ay: Monday Time: 10:00 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

tructions:

a) **Attempt all questions.**
b) **Figures to the right indicate maximum marks.**
c) **Non programmable calculator is allowed.**
d) **Make suitable assumptions wherever necessary.**

a)	Explain the properties of wave function Ψ.	3	CO1 BL
b)	A photon with momentum $p_0 = \frac{E_0}{c} = 1.02\ MeV/c$, scattered with an stationary electron and scattered photon momentum becomes $p_1 = \frac{E_1}{c} = 0.255\ MeV/c$. Find the scattering angle (ϕ) of the photon.	3	CO1 BL
c)	Illuminating the surface of the metal cathode, alternately with light of wavelengths $\lambda_1 = 0.35\ \mu m$ and $\lambda_2 = 0.54\ \mu m$. It was found that corresponding maximum velocities of the photoelectrons differ by a factor $\eta = 2$. Solve the work function of the metal cathode. Where, $c=3.0\times10^8$ m/sec, $m_e=9.1\times10^{-31}$ kg, and $h=6.626\times10^{-34}$ Joules-sec.	3	CO1 BL
d)	A wave function associated with a particle is given by $\Psi = Ae^{-i(\omega t - kx)}$, show that $E\Psi = i\hbar\frac{d\Psi}{dt}$. Where p, ω *and* k are momentum, angular frequency, and wavenumber respectively.	3	CO1 BL
e)	Explain the concept of the Bloch sphere and its role in representing qubit states in quantum computing.	3	CO1 BI

a)	Explain the Fermi energy? If the fermi distribution function is defined by $\boldsymbol{f(E)} = \frac{1}{e^{\frac{(E-E_F)}{k_BT}}+1}$, find the condition for the fermi function $\boldsymbol{f(E) = 1}$	3	CO2 BI
b)	Compare the classical and quantum free electron theories.	3	CO2 BI

Q.3	Explain the Hall Effect and its physical significance, highlighting key applications of this phenomenon in modern technology.	3	C(
Q.4	Explain how laser technologies are utilized in the fields of security and medicine, highlighting their key applications in each domain.	3	C(
Q.5	Explain, how do the AC and DC Josephson effects differ, and in what ways do they appear in superconducting systems.	3	C(

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